Chapter 2
Combinational Logic Design

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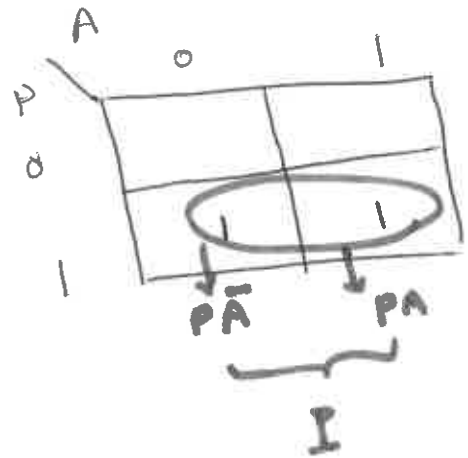
KARNAUGH MAPS

Karnaugh maps (K-maps) are a graphical method for simplifying Boolean equations.

Recall that logic minimization involves combining terms. Two terms containing an implicant, P, and the true and complementary forms of some variable, A, are combined to eliminate A:

$$Y = PA + P\overline{A} = \underset{T_8}{P}(A + \overline{A}) \overset{T_1}{=} P \cdot 1 = P$$

P	A	Y	minterms = T
0	0	0	$\overline{P}\,\overline{A}$
0	1	0	$\overline{P}A$
1	0	1	$P\overline{A}$
1	1	1	PA



sum of products

$Y = P\overline{A} + PA = P$

KARNAUGH MAPS

- Figure 2.43 shows the truth table and K-map for a three-input function.
- The top row of the K-map gives the four possible values for the A and B inputs.
- The left column gives the two possible values for the C input.
- Each square in the K-map corresponds to a row in the truth table and contains the value of the output, Y, for that row.
- Just like each row in a truth table, each square in a K-map represents a single minterm

KARNAUGH MAPS

Each square, or minterm, differs from an adjacent square by a change in a single variable.

This means that adjacent squares share all the same literals except one, which appears in true form in one square and in complementary form in the other.

You may have noticed that the A and B combinations in the top row are in a peculiar order: 00, 01,11, 10. This order is called a Gray code.

The K-map also "wraps around." The squares on the far right are effectively adjacent to the squares on the far left, in that they differ only in one variable, A.

KARNAUGH MAPS

T8 T1

$Y = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C = \bar{A}\bar{B}(\bar{C}+C) = \bar{A}\bar{B}\cdot(1) = \bar{A}\bar{B}$

Sum of Product

$2^3 = 8$

2^x

minterm

Row	A	B	C	Y
0 $\bar{A}\bar{B}\bar{C}$	0	0	0	1
1 $\bar{A}\bar{B}C$	0	0	1	1
2 $\bar{A}B\bar{C}$	0	1	0	0
3 $\bar{A}BC$	0	1	1	0
4 $A\bar{B}\bar{C}$	1	0	0	0
5 $A\bar{B}C$	1	0	1	0
6 $AB\bar{C}$	1	1	0	0
7 ABC	1	1	1	0

(a)

Y, C \ AB	00	01	11	10
0	1	0	0	0
1	1	0	0	0

(b)

Y value

Y, C \ AB	00	01	11	10
0	$\bar{A}\bar{B}\bar{C}$ 0	$\bar{A}B\bar{C}$ 1	$AB\bar{C}$ 3	$A\bar{B}\bar{C}$ 2
1	$\bar{A}\bar{B}C$ 4	$\bar{A}BC$ 5	ABC 7	$A\bar{B}C$ 6

(c)

minterms

Figure 2.43 Three-input function: (a) truth table, (b) K-map, (c) K-map showing minterms

$\bar{A}\bar{B} = Y$

Y, C \ AB	00	01	11	10
0	1	0	0	0
1	1	0	0	0

Figure 2.44 K-map minimization

KARNAUGH MAPS - Circular Thinking

- In the K-map in Figure 2.43, only two minterms are present in the equation as indicated by the 1's in the left column.

$$\overline{A}\overline{B}\overline{C} \text{ and } \overline{A}\overline{B}C$$

- Reading the minterms from the K-map is exactly equivalent to reading equations in sum-of-products form directly from the truth table.

$$Y = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C = \overline{A}\overline{B}(\overline{C} + C) = \overline{A}\overline{B}$$

KARNAUGH MAPS - Circular Thinking

K-maps help us do this simplification graphically by circling 1's in adjacent squares, as shown in Figure 2.44.

For each circle, we write the corresponding implicant. An implicant is the product of one or more literals. Variables whose true and complementary forms are both in the circle are excluded from the implicant

Y is TRUE when A= B=0, independent of C, So the implicant is

$$\overline{A}\,\overline{B}$$

Logic Minimization with K-Maps

- Use the fewest circles necessary to cover all the 1's.
- All the squares in each circle must contain 1's.
- Each circle must span a rectangular block that is a power of 2 (i.e., 1, 2, or 4) squares in each direction.

Logic Minimization with K-Maps

- Each circle should be as large as possible.
- A circle may wrap around the edges of the K-map.
- A 1 in a K-map may be circled multiple times if doing so allows fewer circles to be used.

Example- SEVEN-SEGMENT DISPLAY DECODER

A *seven-segment display decoder* takes a 4-bit data input, $D_{3:0}$, and produces seven outputs to control light-emitting diodes to display a digit from 0 to 9. The seven outputs are often called segments *a* through *g*, or S_a–S_g, as defined in Figure 2.47. The digits are shown in Figure 2.48. Write a truth table for the outputs, and use K-maps to find Boolean equations for outputs S_a and S_b. Assume that illegal input values (10–15) produce a blank readout.

$2^4 = 16$

D_3	D_2	D_0	D_1	Decimal
0	0	0	0	→ 0
0	0	0	1	→ 1
0	0	1	0	→ 2
⋮				
1	0	0	1	→ 9

10–15 → invalid inputs

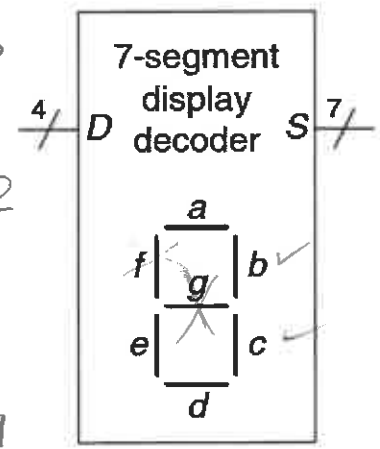


Figure 2.47 Seven-segment display decoder icon

Example- SEVEN-SEGMENT DISPLAY DECODER

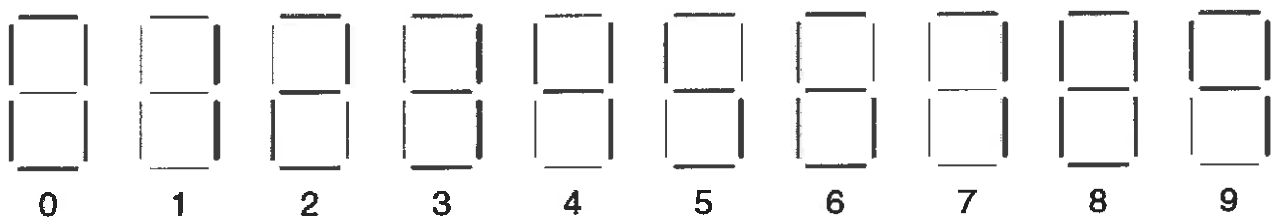


Figure 2.48 Seven-segment display digits

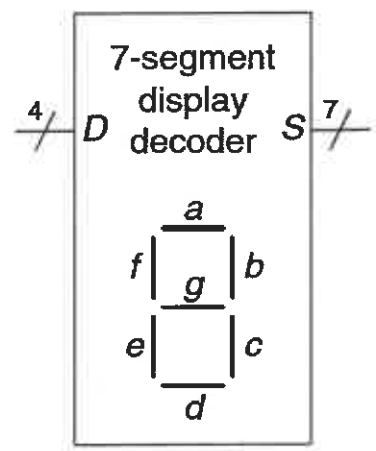
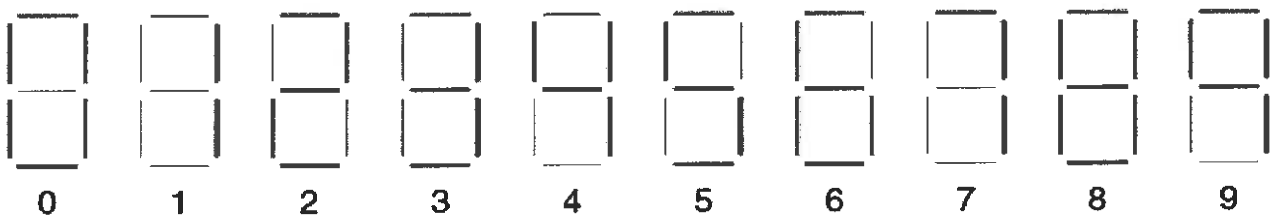


Figure 2.47 Seven-segment display decoder icon

Example- SEVEN-SEGMENT DISPLAY DECODER



BEQ

Decimal value

Table 2.6 Seven-segment display decoder truth table

	$D_{3:0}$	S_a	S_b	S_c	S_d	S_e	S_f	S_g
0	0000	1	1	1	1	1	1	0
1	0001	0	1	1	0	0	0	0
2	0010	1	1	0	1	1	0	1
3	0011	1	1	1	1	0	0	1
4	0100	0	1	1	0	0	1	1
5	0101	1	0	1	1	0	1	1
6	0110	1	0	1	1	1	1	1
7	0111	1	1	1	0	0	0	0
8	1000	1	1	1	1	1	1	1
9	1001	1	1	1	0	0	1	1
10 – 15	others	0	0	0	0	0	0	0

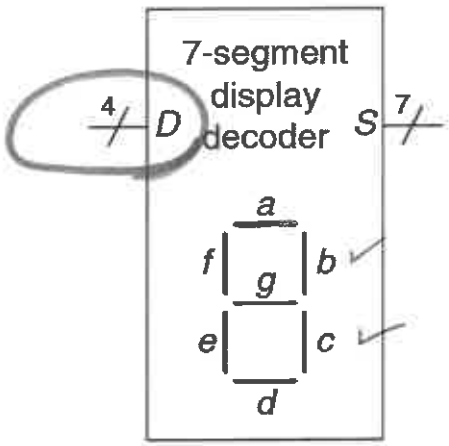


Figure 2.47 Seven-segment display decoder icon

Example- SEVEN-SEGMENT DISPLAY DECODER

	$\overline{D_1}\,\overline{D_0}$	$\overline{D_1}\,D_0$	$D_1 D_0$	$D_1\,\overline{D_0}$
D1D0:	00	01	11	10
D3D2 00	0	1	3	~~4~~ 2
01	~~5~~ 4	~~6~~ 5	~~8~~ 7	~~7~~ 6
11	~~12~~ 12	~~13~~ 13	~~15~~ 15	~~14~~ 14
10	~~9~~ 8	~~10~~ 9	~~12~~ 11	~~11~~ 10

↓ input

4

$2 = 16$

Row#	D3 D2 D1 D0	MIN TERM
0	"0 0 0 0"	$\overline{D3}\,\overline{D2}\,\overline{D1}\,\overline{D0}$
1	"0 0 0 1"	$\overline{D3}\,\overline{D2}\,\overline{D1}\,D0$
2	"0 0 1 0"	$\overline{D3}\,\overline{D2}\,D1\,\overline{D0}$
3	"0 0 1 1"	$\overline{D3}\,\overline{D2}\,D1\,D0$
4	"0 1 0 0"	$\overline{D3}\,D2\,\overline{D1}\,\overline{D0}$
5	"0 1 0 1"	$\overline{D3}\,D2\,\overline{D1}\,D0$
6	"0 1 1 0"	$\overline{D3}\,D2\,D1\,\overline{D0}$
7	"0 1 1 1"	$\overline{D3}\,D2\,D1\,D0$
8	"1 0 0 0"	$D3\,\overline{D2}\,\overline{D1}\,\overline{D0}$
9	"1 0 0 1"	$D3\,\overline{D2}\,\overline{D1}\,D0$
10	"1 0 1 0"	$D3\,\overline{D2}\,D1\,\overline{D0}$
11	"1 0 1 1"	$D3\,\overline{D2}\,D1\,D0$
12	"1 1 0 0"	$D3\,D2\,\overline{D1}\,\overline{D0}$
13	"1 1 0 1"	$D3\,D2\,\overline{D1}\,D0$
14	"1 1 1 0"	$D3\,D2\,D1\,\overline{D0}$
15	"1 1 1 1"	$D3\,D2\,D1\,D0$

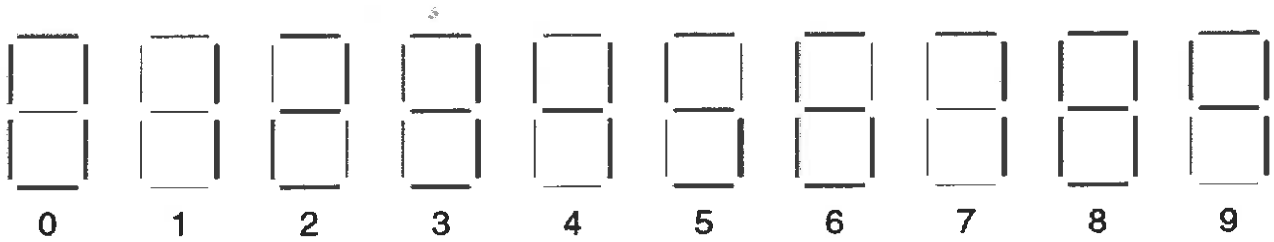
Example- SEVEN-SEGMENT DISPLAY DECODER

	$\overline{D_3}\,\overline{D_2}$	$\overline{D_3}D_2$	D_3D_2	$D_3\overline{D_2}$
D3D2:	00	01	11	10
D1D0				
$\overline{D_1}\,\overline{D_0}$ 00	0	4	12	8
$\overline{D_1}D_0$ 01	1	5	13	9
D_1D_0 11	3	7	15	11
$D_1\overline{D_0}$ 10	2	6	14	10

$2^4 = 16$

Row#	D3	D2	D1	D0	MIN TERM
0	"0	0	0	0"	$\overline{D3}\,\overline{D2}\,\overline{D1}\,\overline{D0}$
1	"0	0	0	1"	$\overline{D3}\,\overline{D2}\,\overline{D1}\,D0$ x
2	"0	0	1	0"	$\overline{D3}\,\overline{D2}\,D1\,\overline{D0}$
3	"0	0	1	1"	$\overline{D3}\,\overline{D2}\,D1\,D0$
4	"0	1	0	0"	$\overline{D3}\,D2\,\overline{D1}\,\overline{D0}$
5	"0	1	0	1"	$\overline{D3}\,D2\,\overline{D1}\,D0$
6	"0	1	1	0"	$\overline{D3}\,D2\,D1\,\overline{D0}$
7	"0	1	1	1"	$\overline{D3}\,D2\,D1\,D0$
8	"1	0	0	0"	$D3\,\overline{D2}\,\overline{D1}\,\overline{D0}$
9	"1	0	0	1"	$D3\,\overline{D2}\,\overline{D1}\,D0$ x
10	"1	0	1	0"	$D3\,\overline{D2}\,D1\,\overline{D0}$
11	"1	0	1	1"	$D3\,\overline{D2}\,D1\,D0$
12	"1	1	0	0"	$D3\,D2\,\overline{D1}\,\overline{D0}$
13	"1	1	0	1"	$D3\,D2\,\overline{D1}\,D0$
14	"1	1	1	0"	$D3\,D2\,D1\,\overline{D0}$
15	"1	1	1	1"	$D3\,D2\,D1\,D0$

Example- SEVEN-SEGMENT DISPLAY DECODER



ROW
Values

Table 2.6 Seven-segment display decoder truth table

	$D_{3:0}$	S_a	S_b	S_c	S_d	S_e	S_f	S_g
0	0000	1	1	1	1	1	1	0
1	0001	0	1	1	0	0	0	0
2	0010	1	1	0	1	1	0	1
3	0011	1	1	1	1	0	0	1
4	0100	0	1	1	0	0	1	1
5	0101	1	0	1	1	0	1	1
6	0110	1	0	1	1	1	1	1
7	0111	1	1	1	0	0	0	0
8	1000	1	1	1	1	1	1	1
9	1001	1	1	1	0	0	1	1
10 - 15	others	0	0	0	0	0	0	0

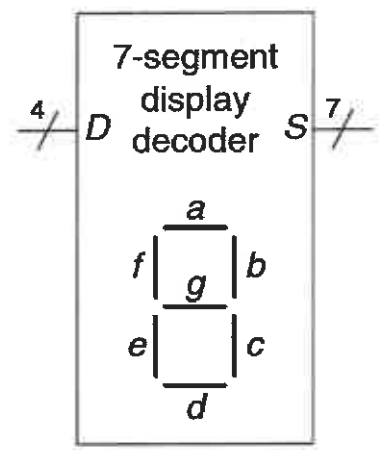


Figure 2.47 Seven-segment display decoder icon

Example- SEVEN-SEGMENT DISPLAY DECODER

$\bar{D}_3\bar{D}_2 \quad \bar{D}_3 D_2 \quad D_3 D_2 \quad D_3\bar{D}_2$

S_a

$D_{1:0}$ \ $D_{3:2}$	00	01	11	10
$\bar{D}_1\bar{D}_0$ (00)	1	0	0	1
$\bar{D}_1 D_0$ (01)	0	1	0	1
$D_1 D_0$ (11)	1	1	0	0
$D_1\bar{D}_0$ (10)	1	1	0	0

S_b

$D_{1:0}$ \ $D_{3:2}$	00	01	11	10
00	1	1	0	1
01	1	0	0	1
11	1	1	0	0
10	1	0	0	0

$$S_a = \bar{D}_3 D_1 + \bar{D}_3 D_2 D_0 + D_3 \bar{D}_2 \bar{D}_1 + \bar{D}_2 \bar{D}_1 \bar{D}_0$$

$$S_b = \bar{D}_3 \bar{D}_2 + \bar{D}_2 \bar{D}_1 + \bar{D}_3 \bar{D}_1 \bar{D}_0 + \bar{D}_3 D_1 D_0$$

Example- SEVEN-SEGMENT DISPLAY DECODER

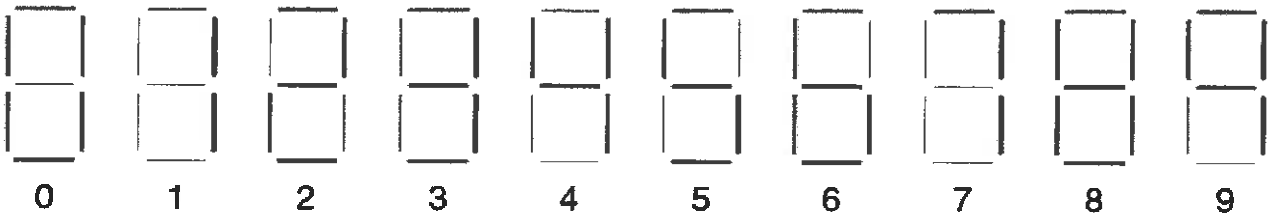
S_a ($D_{3:2}$ \ $D_{1:0}$)	00	01	11	10
00	1	0	0	1
01	0	1	0	1
11	1	1	0	0
10	1	1	0	0

S_a ($D_{3:2}$ \ $D_{1:0}$)	00	01	11	10
00	1	0	0	1
01	0	1	0	1
11	1	1	0	0
10	1	1	0	0

$$S_a = \bar{D}_3 D_1 + \bar{D}_3 D_2 D_0 + D_3 \bar{D}_2 \bar{D}_1 + \bar{D}_2 \bar{D}_1 \bar{D}_0$$

$$S_a = \bar{D}_3 D_1 + \bar{D}_3 D_2 D_0 + D_3 \bar{D}_2 \bar{D}_1 + \bar{D}_3 \bar{D}_2 \bar{D}_0$$

Example- SEVEN-SEGMENT DISPLAY DECODER



Input

Table 2.6 Seven-segment display decoder truth table

	$D_{3:0}$	S_a	S_b	S_c	S_d	S_e	S_f	S_g
0	0000	1	1	1	1	1	1	0
1	0001	0	1	1	0	0	0	0
2	0010	1	1	0	1	1	0	1
3	0011	1	1	1	1	0	0	1
4	0100	0	1	1	0	0	1	1
5	0101	1	0	1	1	0	1	1
6	0110	1	0	1	1	1	1	1
7	0111	1	1	1	0	0	0	0
8	1000	1	1	1	1	1	1	1
9	1001	1	1	1	0	0	1	1
10 – 15	others	X	X	X	X	X	X	X

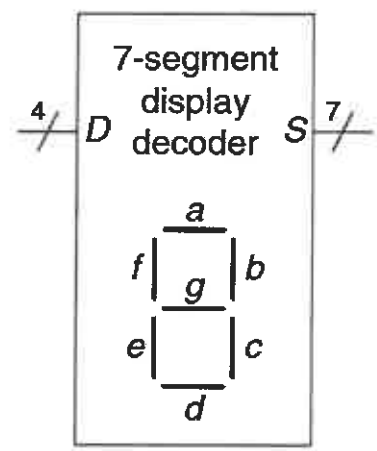


Figure 2.47 Seven-segment display decoder icon

Repeat Example 2.10 if we don't care about the output values for illegal input values of 10 to 15.

Example- SEVEN-SEGMENT DISPLAY DECODER

incorrect circles why?

S_a \ $D_{3:2}$, $D_{1:0}$	00	01	11	10
00	1	0	0	1
01	0	1	0	1
11	1	1	0	0
10	1	1	0	0

$S_b =$

10 – 15

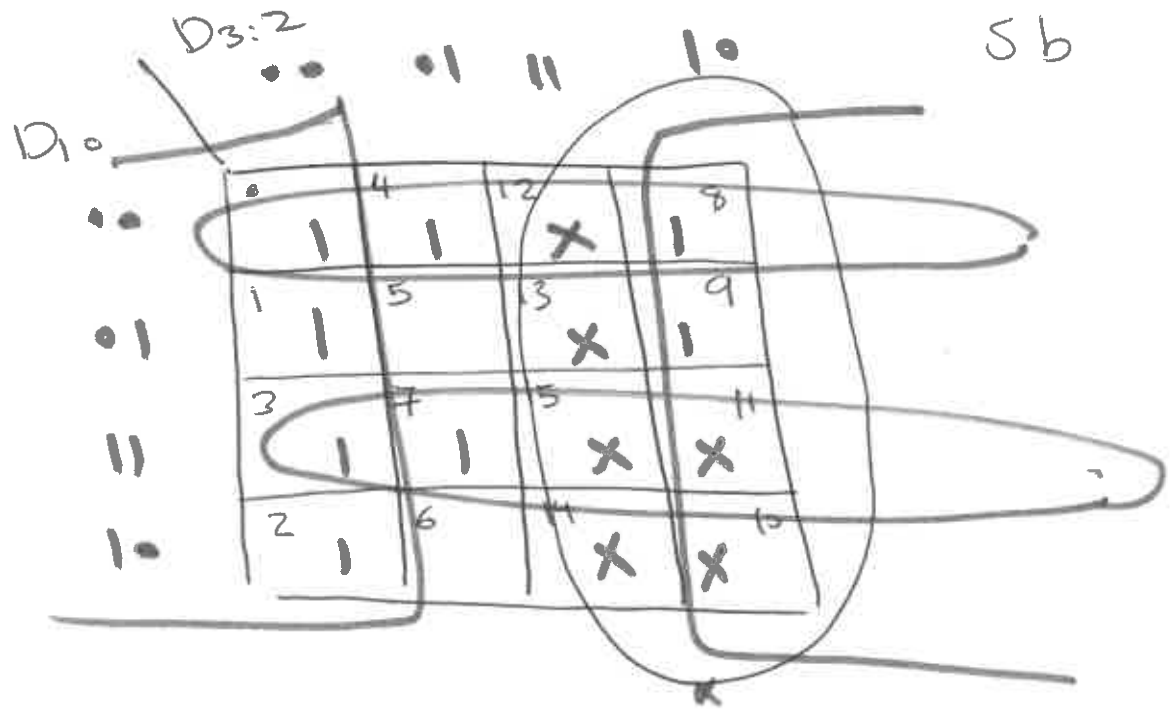
S_b \ $D_{3:2}$, $D_{1:0}$	00	01	11	10
00	1	1	X	1
01	1	0	X	1
11	1	1	X	X
10	1	0	X	X

An implicant is called prime if it can not be combined with other implicant to form a new implicant

$$S_a = \bar{D}_3 D_1 + \bar{D}_3 D_2 D_1 + D_3 \bar{D}_2 \bar{D}_1 + \underbrace{\bar{D}_3 \bar{D}_2 \bar{D}_1 \bar{D}_0}_{\text{prime implicant}}$$

prime imp.

Example- SEVEN-SEGMENT DISPLAY DECODER



$$S_b = D_3 + D_2 + D_1 D_0 + \bar{D}_1 \bar{D}_0$$

Example- SEVEN-SEGMENT DISPLAY DECODER